

Manan Pal

+91-9635211275 · mananpal27@gmail.com · [LinkedIn](#) · [GitHub](#) · [Portfolio](#)

Summary

Final-year B.Tech CSE student at KIIT University (CGPA: 8.51/10) with end-to-end project experience in ML, deep learning, NLP, and graph neural networks. Proficient in Python, TensorFlow, PyTorch, and Scikit-learn; strong foundations in DSA, OOP, and DBMS.

Education

KIIT University — B.Tech, Computer Science and Engineering

May 2023 – May 2027

CGPA: 8.51 / 10

Technical Skills

Languages: Python, Java, C, SQL, JavaScript

AI/ML: Machine Learning, Deep Learning, NLP, Computer Vision, Graph Neural Networks, Generative AI

Frameworks & Libraries: TensorFlow, Keras, PyTorch, Scikit-learn, Pandas, NumPy, PyTorch Geometric, NetworkX

Tools & Platforms: Git, GitHub, REST APIs, Streamlit, Jupyter Notebook, Google Colab, VS Code, Vercel

Core CS: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Computer Networks

Projects

Cybersecurity Attack Path Prediction using GNN | *Python · PyTorch · GraphSAGE · GCN · Streamlit* [GitHub](#)

- Designed a heterogeneous synthetic graph dataset modeling network hosts, traffic flows, and node-level threat features to simulate real attack topologies.
- Architected and trained GCN and GraphSAGE models for node-level threat classification and multi-hop attack path prediction using PyTorch Geometric.
- Developed a full inference pipeline with graph visualization and a Streamlit dashboard enabling interactive threat analysis and prediction inspection.

Brain Tumor Detection: CNN vs ViT with GAN Augmentation | *TensorFlow · MobileNetV2 · Vision Transformer · Streamlit* [GitHub](#)

- Designed and trained MobileNetV2 and Vision Transformer (ViT) models for multi-class brain tumor classification on MRI data; ViT achieved 99.8% accuracy.
- Integrated GAN-based synthetic MRI generation for class-balanced augmentation, Grad-CAM saliency mapping for explainability, and NLP-based symptom triage.
- Deployed the full multi-model pipeline as an interactive Streamlit application with per-model comparison and explainability visualization.

NLP-Based Resume Ranking System | *Python · TF-IDF · Scikit-learn · Streamlit* [GitHub](#)

- Engineered a resume screening pipeline that extracts text from PDF uploads and ranks candidates against job descriptions via TF-IDF cosine similarity scoring.
- Implemented keyword extraction and skill-gap analysis to surface matched and missing competencies per candidate; deployed via Streamlit with batch PDF upload and exportable ranked output.

Student Placement Readiness Prediction | *Scikit-learn · Random Forest · Tkinter* [GitHub](#)

- Benchmarked Logistic Regression, Decision Tree, SVM, and Random Forest classifiers on academic and extracurricular feature sets for placement prediction.
- Achieved 89.5% accuracy with Random Forest; packaged the trained model into a Tkinter desktop application with probability-based output and feature importance plots.

Developer Portfolio Website | *Next.js · TypeScript · Tailwind CSS · Vercel*

- Built and deployed a responsive portfolio site using Next.js and Tailwind CSS, showcasing AI/ML projects with links to live deployments and GitHub repositories.
- Implemented responsive component architecture with optimized layout and cross-device compatibility; hosted on Vercel with continuous deployment.

Virtual Experience

AICTE – EduSkills

2025

AI/ML (AWS Academy) · Android Development (Google) · Generative AI Fundamentals

Certifications

Machine Learning Foundations — AWS Academy

AWS Cloud Foundations — AWS Academy

Introduction to Software Engineering — IBM, Coursera

Ethical Decision Making in Tech — Univ. of Colorado Boulder

Business for Good — London Business School